

Uncertainty Calibration of Multi-Label Bird Sound Classifiers

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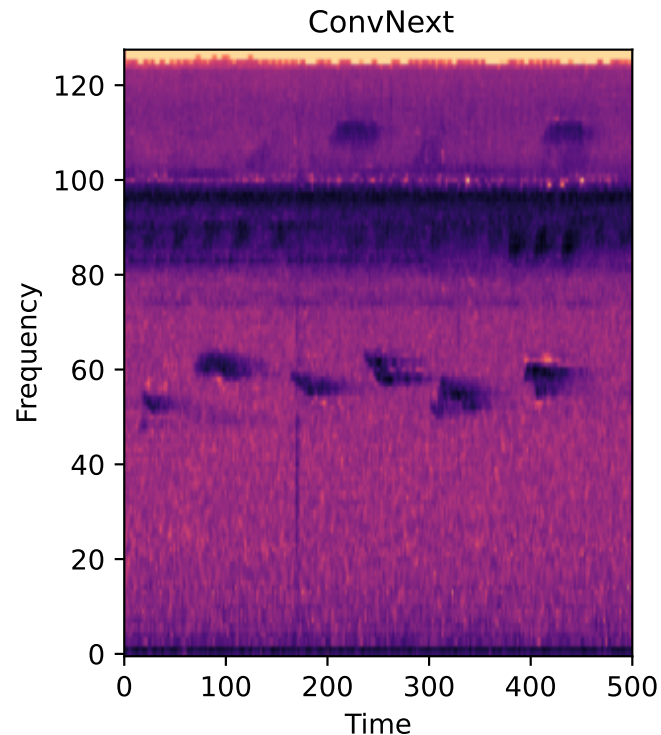
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Motivation

- **Passive acoustic monitoring (PAM)** enables large-scale biodiversity assessment ^[1]
- Reliable decisions need **well-calibrated uncertainty** — not only high accuracy ^[2]
- **Overconfidence** → biased occupancy, missed human review
- **Underconfidence** → scores less useful for prioritisation
- **Challenges:** multi-label (overlapping vocalisations), long-tailed species, distribution shift train vs. deployment ^[3]



Spectrogram of a 5s snippet of the PER PAM dataset

[1] Sugai et al., *Terrestrial Passive Acoustic Monitoring: Review and Perspectives*, *BioScience* 2019

[2] Guo et al., *On Calibration of Modern Neural Networks*, *ICML* 2017

[3] Stowell et al., *Computational Bioacoustics with Deep Learning: A Review and Roadmap*, *PeerJ* 2022

Contributions

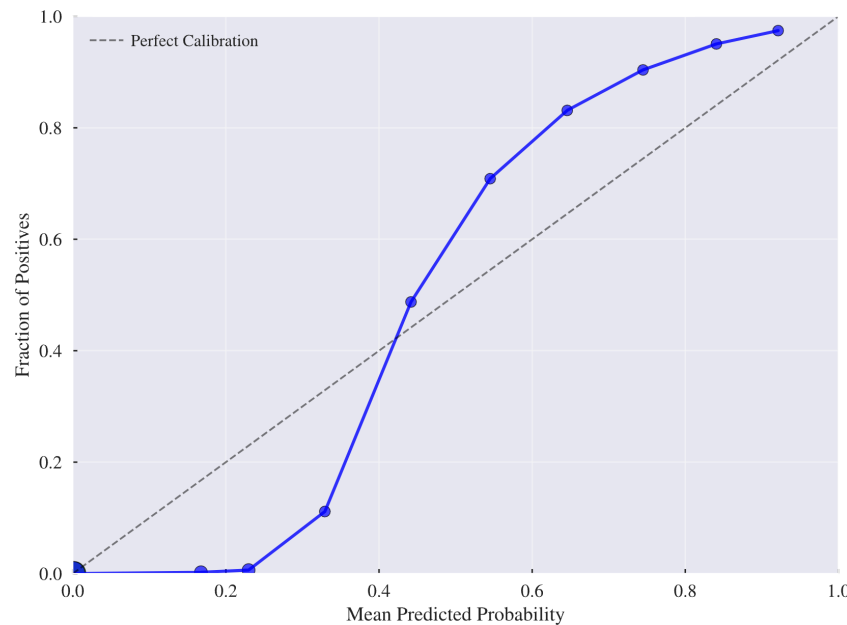
Contributions

- C1. Benchmark calibration** of four state-of-the-art multi-label bird sound classifiers on BirdSet, reporting global, per-dataset and per-class calibration with threshold-free metrics, revealing that per-domain (e.g., dataset, class, etc.) calibration is highly variable.
- C2. Investigate post hoc strategies** (temperature and Platt scaling) showing that Platt scaling with 10 minutes of test data can significantly improve calibration.
- C3. Summarise implications** to give practitioners concrete suggestions for deployment-specific calibration evaluation and improvement. Code is available online.^a

^a <https://github.com/DBD-research-group/UncertainBird>

Reliability Diagram

- Predictions are binned by confidence; bar height = observed positive rate in that bin
- **Diagonal line** = perfect calibration (confidence matches reality)
- **Below diagonal** = overconfident (model predicts higher than reality)
- **Above diagonal** = underconfident (model predicts lower than reality)



Example reliability diagram.

Calibration Metrics: ECE and MCS

ECE (Expected Calibration Error) — magnitude of miscalibration (lower is better):

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$$

- Does not reveal *direction* (over- vs underconfident)

MCS (Miscalibration Score) ^[1] — signed version reveals direction:

$$\text{MCS} = \sum_{m=1}^M \frac{|B_m|}{N} (\text{conf}(B_m) - \text{acc}(B_m))$$

- MCS = 0: perfectly calibrated · MCS > 0: overconfident · MCS < 0: underconfident
- Caveat: over- and underconfident bins can cancel each other out

OCS and UCS: Decomposing Miscalibration

OCS (Overconfidence Score) — only positive deviations:

$$\text{OCS} = \sum_{m=1}^M \frac{|B_m|}{N} \max(\text{conf}(B_m) - \text{acc}(B_m), 0)$$

UCS (Underconfidence Score) — only negative deviations:

$$\text{UCS} = \sum_{m=1}^M \frac{|B_m|}{N} |\min(\text{conf}(B_m) - \text{acc}(B_m), 0)|$$

- Together they give a complete picture: a model can be overconfident in some bins and underconfident in others
- **ECE** = OCS + UCS · **MCS** = OCS - UCS

Experimental Setup

Models: 4 SoTA multi-label bird sound classifiers

Dataset:

Model	Architecture	Params	
<i>AudioProtoPNet</i> – 20 ^[1]	ConvNeXt + prototypes	287M	■ BirdSet benchmark ^[3] ■ Train: XCL (~7,200 h, 9,734 classes) ■ Val: POW (6.3 h) ■ Test: PER, NES, UHH, HSN, NBP, SSW, SNE (7 PAM datasets, expert-annotated) ■ 5s windows and 32 kHz
<i>BirdMAE</i> ^[2]	ViT-B, SSL+SL	92.9M	
<i>ConvNeXt_{BS}</i> ^[3]	ConvNeXt, SL	88M	
<i>Perch v2</i> ^[4]	EfficientNet-B3, SSL+SL	12M	

[1] Heinrich et al., *AudioProtoPNet: An Interpretable Deep Learning Model for Bird Sound Classification*, *Ecological Informatics* 2025

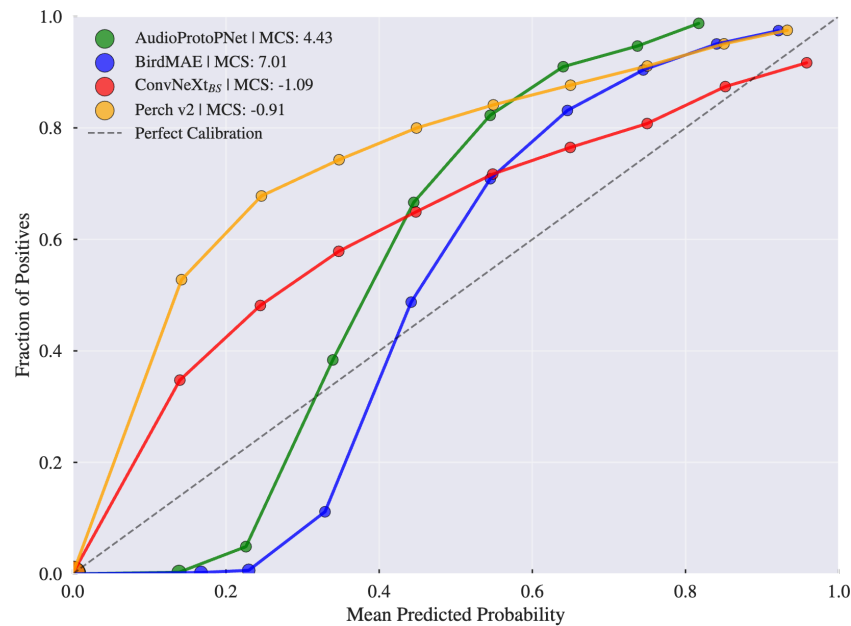
[2] Rauch et al., *Can Masked Autoencoders Also Listen to Birds?*, *TLMR* 2025

[3] Rauch et al., *BirdSet: A Large-Scale Dataset for Audio Classification in Avian Bioacoustics*, *ICLR* 2025

[4] van Merriënboer et al., *Perch 2.0: The Bittern Lesson for Bioacoustics*, *ArXiv preprint* 2025

Global Calibration Results

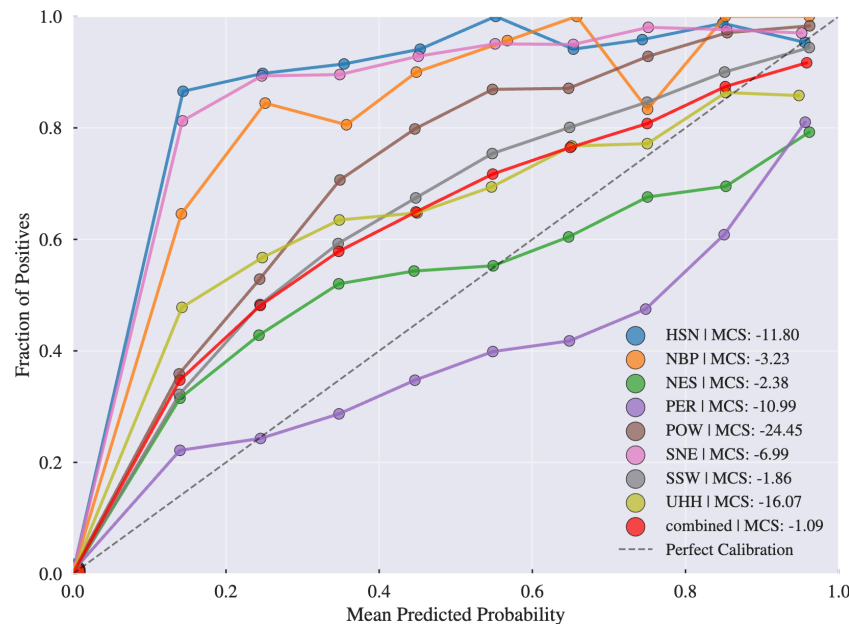
- *Perch v2* and *ConvNeXt_{BS}*: best global calibration ($MCS \approx -1$), slightly **underconfident**
- *AudioProtoPNet* and *BirdMAE*: **overconfident** (MCS 4.4, 7.0)
- **Best cmAP $\neq$ best calibration** — *AudioProtoPNet* leads on discrimination but is poorly calibrated



Reliability diagrams computed over all test segments.

Per-Dataset Calibration

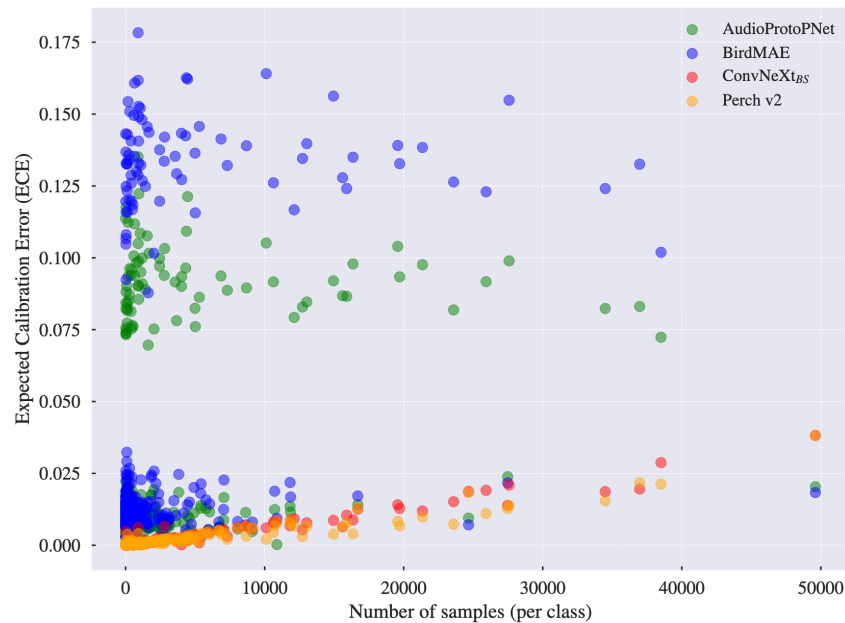
- **Strong variability** across datasets (e.g. *Perch v2* ECE from 1.37 to 22.46)
- **Aggregated ECE** across all test data can **mask** domain-specific miscalibration
- *Perch v2* & *ConvNeXt_{BS}*: underconfident on most sets; *AudioProtoPNet* & *BirdMAE* overconfident except on some (e.g. POW, UHH)



ConvNeXt_{BS} reliability per BirdSet dataset. Large variability across domains.

Per-Class Calibration — ECE vs. Sample Count

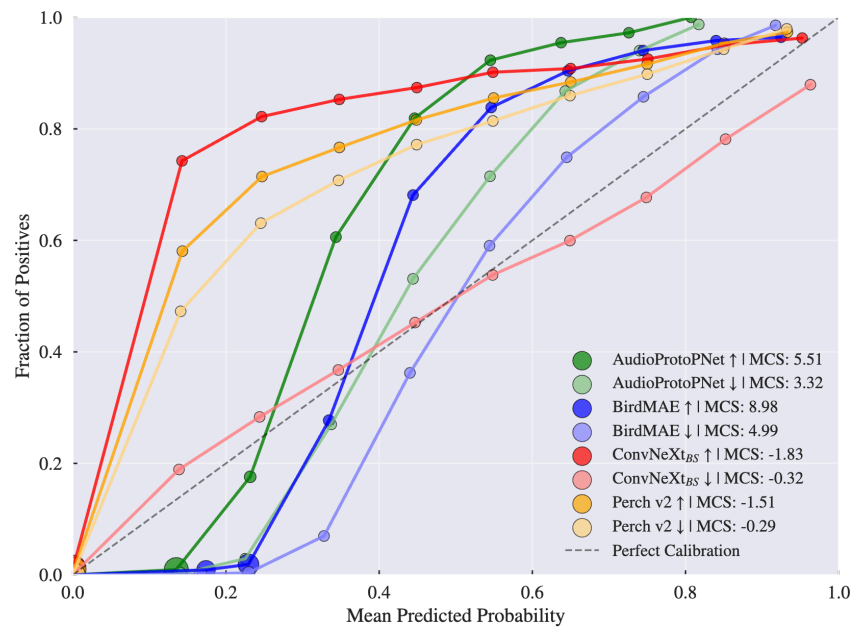
- *Perch v2* & *ConvNeXt_{BS}*: **low ECE** that increases **linearly** with sample count
- *AudioProtoPNet* & *BirdMAE*: no clear structure
- For the **largest** classes, *AudioProtoPNet* / *BirdMAE* eventually reach lower ECE than *Perch v2* / *ConvNeXt_{BS}*
- **Wide ECE spread** for underrepresented classes in *AudioProtoPNet* / *BirdMAE* explains their poor per-dataset scores — poorly calibrated classes can mask well-calibrated ones



Class-wise ECE over number of samples per class.

Per-Class Calibration — Common vs. Rare Species

- Rare classes generally have less positives per probability bin.
- **All models show lower MCS for rare classes.**
- *ConvNeXt_{BS}* has a particular strong improvement.



Reliability diagrams for common (↑) and rare (↓) species subsets.

Calibration Methods

Post hoc — learn a mapping from logits to calibrated probabilities, no retraining:

- **Temperature Scaling (TS)** ^[1] — single scalar T rescales logits
- **Platt Scaling (PS)** ^[2] — adds a bias b , corrects offset errors
- Isotonic regression ^[3], vector scaling ^[4] — more flexible variants

Bayesian / ensemble — principled but computationally expensive:

- MC Dropout ^[5]
- Deep Ensembles ^[6]

Ad hoc — bake calibration into training (label smoothing ^[7], focal loss ^[8], mixup ^[9])

→ **We focus on TS and PS**: simple, parameter-efficient, no retraining required, and well-suited for limited calibration data ^[1] ^[2]

[1] Guo et al., On Calibration of Modern Neural Networks, ICML 2017

[2] Platt et al., Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods, Advances in Large Margin Classifiers 1999

[3] Niculescu-Mizil et al., Predicting good probabilities with supervised learning, ICML 2005

[4] Kull et al., Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with Dirichlet calibration, NeurIPS 2019

[5] Gal et al., Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning, ICML 2016

[6] Lakshminarayanan et al., Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles, Advances in Neural Information Proces... 2017

[7] Müller et al., When does label smoothing help?, NeurIPS 2019

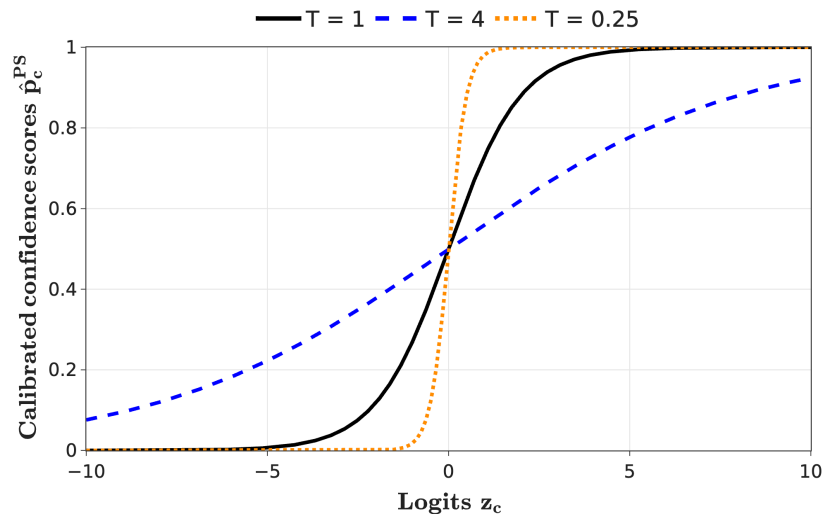
[8] Lin et al., Focal Loss for Dense Object Detection, ICCV 2017

[9] Zhang et al., mixup: Beyond Empirical Risk Minimization, ICLR 2018

Temperature Scaling

$$\hat{p}_c^{\text{TS}} = \sigma(z_c/T), \quad T > 0$$

- $T = 1$: uncalibrated baseline
- $T > 1$: smooths distribution $\rightarrow$ reduces overconfidence
- $T < 1$: sharpens distribution $\rightarrow$ reduces underconfidence
- Preserves monotonic ranking of scores when T is global ^[1]
- **Limitation:** $\sigma(0) = 0.5 \rightarrow$ cannot correct systematic offset errors ^[2]



Temperature scaling — effect of T on $\sigma(z/T)$.

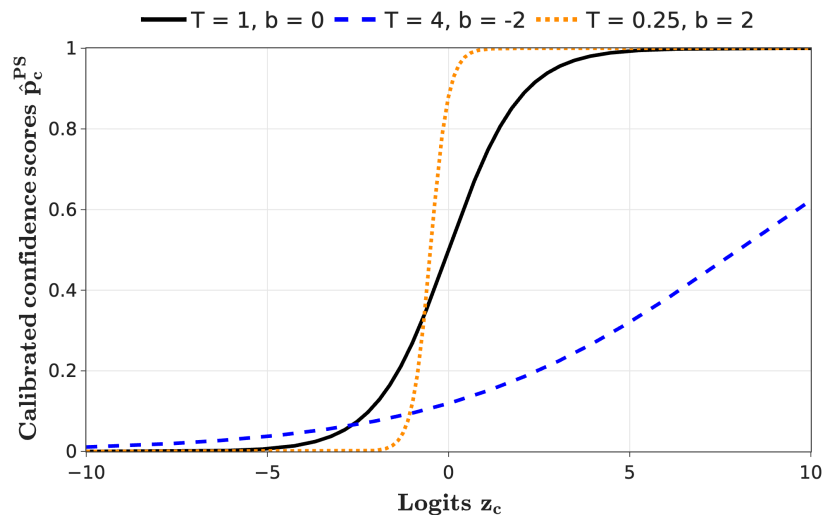
[1] Guo et al., On Calibration of Modern Neural Networks, ICML 2017

[2] Alexandari et al., Maximum Likelihood with Bias-Corrected Calibration is Hard-To-Beat at Label Shift Adaptation, ICML 2020

Platt Scaling

$$\hat{p}_c^{\text{PS}} = \sigma(z_c/T + b), \quad T > 0, b \in \mathbb{R}$$

- **Bias b :** shifts the sigmoid left/right
 - $b > 0$: increases confidence (corrects underconfidence)
 - $b < 0$: decreases confidence (corrects overconfidence)
- Overcomes TS offset limitation [1]
- **Per-class** variant: fit (T_c, b_c) per class [2]
- Parameters fitted by minimising NLL on held-out data [3] [4]



Platt scaling — temperature T and bias b . $\sigma(z/T + b)$.

- [1] Alexandari et al., Maximum Likelihood with Bias-Corrected Calibration is Hard-To-Beat at Label Shift Adaptation, ICML 2020
- [2] Frenkel et al., Network calibration by class-based temperature scaling, 2021 29th European Signal Processing ...
- [3] Lin et al., A note on Platt's probabilistic outputs for support vector machines, Machine learning 2007
- [4] Guo et al., On Calibration of Modern Neural Networks, ICML 2017

Post Hoc Calibration: Experimental Setting

Two calibration-data settings:

1. **Global** — fit single (T) or (T, b) on **POW** validation set
 - POW classes differ from test sets → only global params
2. **Per-class** — fit (T_c) or (T_c, b_c) on **first 10 min** of each test set
 - Enables class-level correction; test set slightly reduced for evaluation

Optimisation: Adam ^[1], lr = 0.001, 1000 steps.

Post Hoc Calibration: Results

Model	TS POW	PS POW	TS/cl.	PS/cl.
AudioProtoPNet	56%	56%	75%	92%
BirdMAE	81%	89%	94%	95%
ConvNeXtBS	10%	20%	67%	12%
Perch v2	33%	24%	23%	61%

Median % MCS improvement over base (all test sets).

- **Per-class PS** gives largest, most consistent gains (AudioProtoPNet ~92%, BirdMAE ~95%)
- Global TS/PS works well for BirdMAE but degrades on some datasets for AudioProtoPNet (e.g. UHH, HSN)
- ConvNeXtBS benefits most from **per-class TS**; Perch v2 from **per-class PS**
- No single method best — improvements are model- and domain-specific

Conclusions & Implications

1. **Calibration varies** across datasets and classes; aggregate metrics can hide systematic miscalibration.
2. **Perch v2 / ConvNeXtBS** consistently underconfident; **AudioProtoPNet / BirdMAE** mixed, with overconfidence at low probabilities.
3. **Simple post hoc** (Platt scaling, 10 min labelled data) can **strongly improve** deployment-specific calibration; effectiveness is model- and domain-dependent.

For deployment: evaluate per-dataset and per-class; report OCS/UCS; consider per-class or subgroup-aware calibration for rare species.

Future: Bayesian/distance-aware methods, calibration@k, more deployment scenarios.

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Takeaways

- **Calibration varies across datasets/classes;**
aggregated metrics hide domain-specific failures
- Perch v2 / ConvNeXt_{BS} are underconfident;
AudioProtoPNet / BirdMAE overconfident
- **Per-class Platt scaling (10 min of data)**
strongly improves calibration



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Code: github.com/DBD-research-group/UncertainBird